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# Optimization of electricity futures and LNG futures trading under practical constraints of power producers in Japan

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**Abstract**

In this study, we developed a mixed-integer programming model that can help maximize the profit of the gas-fired power generation business in Japan. Owing to the practical constraints of liquefied natural gas (LNG) cargo, LNG tank, and power generating unit, Japanese gas-fired power producers are currently unable to effectively utilize the futures market. By using this model, power producers can calculate the optimal futures position and the number of spot LNG cargoes to buy/sell in response to the daily changing forward spark spread. The numerical simulation shows that the optimal electricity futures position decreases in a step-wise manner against the electricity futures price.

**Keywords** power producer, electricity futures, LNG futures, spark spread, mixed-integer programming model

**Research Activity Group** Mathematical Finance

## 1. Introduction

With the full liberalization of Japan's retail electricity market in 2016, the electricity business made a rapid shift to become competitive under the market mechanism. Electricity utilities changed their business model from minimizing the cost of power generation based on expected retail demand to a new business model that seeks to maximize profit based on market trading. Electricity utilities are now faced with the need to hedge their market risk.

This study focuses on gas-fired power generation, which accounts for about 34% of Japan's electricity demand in FY2021 and contributes significantly to ensuring a stable power supply. The gas-fired generation plant is generally regarded as an option whose underlying asset is the spark spread [1, Ch. 9]. Its market risk can be hedged by trading the forward spark spread, which is calculated from the electricity futures price, the gas futures prices, and the heat efficiency of the generating unit. By fixing the forward spark spread, the profit from the gas-fired power generation business can be locked in. Endo and Hattori [2,3] investigated the methods and practices of European electricity utilities to hedge market risk through forward spark spread trading.

For gas-fired power producers, the minimum size of the actual gas transaction is an important consideration. When natural gas is purchased through pipelines, as is the case in Europe and the U.S., power producers can vary the volume of natural gas in small increments. They can trade natural gas when they want to buy/sell, without delay, and in the quantity they need. This gives them the flexibility to hedge their market risk.

However, when natural gas is traded in the form of liquefied natural gas (LNG), as is the case in Japan, natural gas is procured in bulk with LNG cargoes of over 60,000 tons, with a time lag of about 20 days after the decision, and stored in LNG tanks that have operational upper and lower capacity limits [4]. In addition, multiple generating units with different heat efficiencies are connected to the tank.

These practical constraints make it difficult to hedge market risk with fine granularity through forward spark spread trading. Therefore, Japanese gas-fired power producers are currently unable to use the futures market effectively. As Endo and Matsumoto [5] show, the Japanese electricity futures market remains extremely illiquid compared to the European market.

To overcome these difficulties, this study constructs a mixed-integer programming model to maximize the profit of forward spark spread trading, taking into account the practical constraints of LNG cargo, LNG tank, and power generating unit, as described above.

In a preceding study on mathematical programming of plant operation for the power generation business, Shiina [6] developed a stochastic programming model that provides the optimal schedule of power generation units and the generation level of each unit by minimizing the

generation cost while considering the uncertainty of retail demand in Japan.

## 2. Constraints for power producers

### 2.1 Minimum and maximum electricity futures position

We consider $N$ gas-fired generating units and assume that the heat efficiency of each unit is $\alpha_i, i = 1, \ldots, N$. The heat efficiency generally varies depending on the actual operating pattern of the unit at delivery month $T$, but we hold it constant for ease of calculation. In unit $i$, we can set the amount of LNG required to produce 1 GWh of electricity as $L_i$ (tons/GWh) by calculating from its heat efficiency $\alpha_i$.

Japanese power producers generally determine their power generation plan in advance to meet the demand for bilateral wholesale power contracts, and most of the LNG needed for power generation is purchased in advance under long-term contracts [4]. Therefore, we assume that power producers trade electricity futures for increases or decreases from the predetermined generation plan.

In unit $i$, we denote the electricity that can be sold in the electricity futures market by increasing generation as $D_i(t,T) \leq 0$ (GWh), and the electricity that can be bought in the electricity futures market by replacing the predetermined generation plan as $U_i(t,T) \geq 0$ (GWh). In other words, $D_i(t,T)$ and $U_i(t,T)$ are the minimum and maximum position in the electricity futures market, respectively. Let $x_i(t,T)$ be the electricity futures position that power producers take:

$$D_i(t,T) \leq x_i(t,T) \leq U_i(t,T).$$

### 2.2 Number of tradable LNG cargoes

In practice, LNG is transported on a per LNG cargo basis, therefore any additional LNG can only be procured or sold by the bulk of LNG cargo. We denote the standard LNG cargo size as $C$ (tons), and a buffer for volume fluctuations in LNG trading as $B$ (tons). The actual volume in LNG trading is often less or more than the standard cargo size.

As a result of LNG trading, the remaining LNG volume in the tank cannot exceed the maximum/minimum operational capacity limit. This is because LNG terminals in Japan are not interconnected, and the operational capacity limits of each tank are often not very large [4].

We denote the number of spot LNG cargoes that can be traded after taking into account the upper and lower tank capacity limits as $j$, $j = -l, \ldots, 0, \ldots, k$. Here, $l$ is the maximum number of sellable LNG cargoes. $l$ depends mainly on the flexibility of long-term LNG contracts and the minimum capacity limit of the LNG tank. $k$ is the maximum number of buyable spot LNG cargoes. $k$ depends mainly on the liquidity of the spot LNG market and the maximum capacity limit of the LNG tank. In general, $l$ and $k$ vary with the delivery month $T$, but we fix them as constants for ease of calculation.

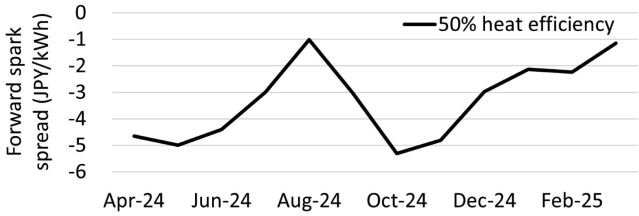


Fig. 1. Forward spark spread calculated from TOCOM West Area Baseload electricity futures and NYMEX Japan/Korea Marker (Platts) LNG futures.

## 3. Forward spark spread trading

We denote the electricity futures price of delivery month $T$ at time $t$ as $P(t,T)$ (JPY/kWh) and the LNG futures price as $F(t,T)$ (USD/MMBTU converted to JPY/kWh). The forward spark spread $S_i(t,T)$ (JPY/kWh), which shows the economic value of the unit $i$ at delivery month $T$, is defined as:

$$S_i(t,T) = P(t,T) - \frac{1}{\alpha_i} F(t,T).$$

Fig. 1 shows the forward spark spread for a 50% heat efficiency generating unit as of September 30, 2023, for delivery month FY2024.

If $S_i(t,T) \geq 0$, power producers can make profit of $-S_i(t,T)x_i(t,T)$ by selling the forward spark spread, which is a combination of selling $x_i(t,T)$ of electricity futures and buying $-L_i x_i(t,T)$ of LNG futures. When the futures delivery month arrives, power producers buy the same amount of spot LNG (whose price is fixed by the LNG futures contract), generate electricity and sell it in the spot electricity market (whose price is fixed by the electricity futures contract).

On the contrary, if $S_i(t,T) < 0$, power producers can make a profit of $-S_i(t,T)x_i(t,T)$ by buying the forward spark spread, which is a combination of buying $x_i(t,T)$ of electricity futures and selling $-L_i x_i(t,T)$ of LNG futures. When the futures delivery month arrives, power producers buy the same amount of electricity in the spot electricity market (whose price is fixed by the electricity futures contract), replace part of the predetermined generation plan with the spot electricity, and resell the LNG in the spot LNG market (whose price is fixed by the LNG futures contract).

## 4. Optimization model

Profit maximization in forward spark spread trading under practical constraints can be formulated as a mixed-integer programming model under disjunctive constraints as follows:

$$\begin{aligned}
\max \; & -\sum_{i=1}^{N} S_i x_i \\
\text{s.t.} \; & -\sum_{i=1}^{N} L_i x_i \leq kC + B + M(1 - y_k), \\
& -\sum_{i=1}^{N} L_i x_i \geq kC - B - M(1 - y_k),
\end{aligned}$$

$$\vdots$$

$$-\sum_{i=1}^{N} L_i x_i \leq B + M(1 - y_0),$$

$$-\sum_{i=1}^{N} L_i x_i \geq -B - M(1 - y_0),$$

$$\vdots$$

$$-\sum_{i=1}^{N} L_i x_i \leq -lC + B + M(1 - y_{-l}),$$

$$-\sum_{i=1}^{N} L_i x_i \geq -lC - B - M(1 - y_{-l}),$$

$$D_i \leq x_i \leq U_i, \quad i = 1, \ldots, N,$$

$$\sum_{j=-l}^{k} y_j = 1,$$

$$y_j \in \{0, 1\}, \quad j = -l, \ldots, 0, \ldots, k.$$

Here, $y_j$ is a binary. If $y_j = 1$, power producers will additionally purchase $j$ number of LNG cargoes from the spot market (if $j$ is negative, power producers will resell the long-term contract LNG cargoes). $M$ is a sufficiently large fixed number.

The objective function is the total profit earned from each unit's forward spark spread trading. Constraints are set to ensure that the total amount of LNG additionally required (or unnecessary and reduced) falls as a multiple of the LNG cargo size $C$. For example, in the case of $y_3 = 1$, $M(1 - y_3) = 0$, the effective constraints are only $3C - B \leq -\sum_{i=1}^{N} L_i x_i \leq 3C + B$, which means that the total amount of LNG required in the optimal solution is approximately three cargoes of LNG. As for the remaining constraints, since $y_j = 0$ for $j \neq 3$ and $M$ is sufficiently large, these constraints are satisfied regardless of $x_i$.

## 5. Numerical simulation

### 5.1 Simulation method and assumptions

We consider three gas-fired power generating units with different heat efficiencies as shown in Table 1.

Unit 1 has the highest heat efficiency and is assumed to operate at its maximum generating capacity to meet demand at minimum cost. The remaining additional sellable electricity is zero ($D_1 = 0$ GWh), while the reducible electricity is large ($U_1 = 800$ GWh). Unit 2 has medium heat efficiency, and the remaining additional/reducible electricity is moderate ($D_2 = -400$ GWh, $U_2 = 400$ GWh). Unit 3 has the lowest heat efficiency and is assumed to operate at a minimum generation capacity. The remaining additional sellable electricity is large ($D_3 = -800$ GWh), while the reducible electricity is zero ($U_3 = 0$ GWh).

The constants are set as follows: LNG cargo size $C = 60,000$ (tons), buffer for volume fluctuations in LNG trading $B = 1,000$ (tons), maximum number of sellable LNG cargoes $l = 2$, maximum number of buyable LNG cargoes $k = 3$, and the currency exchange rate is 145 (JPY/USD).

Table 1. Factors of each power generating unit.

| | $\alpha_i$ | $L_i$ | $D_i$ | $U_i$ |
|---|---|---|---|---|
| Unit 1 | 50% | 132 | 0 | 800 |
| Unit 2 | 40% | 165 | −400 | 400 |
| Unit 3 | 30% | 220 | −800 | 0 |

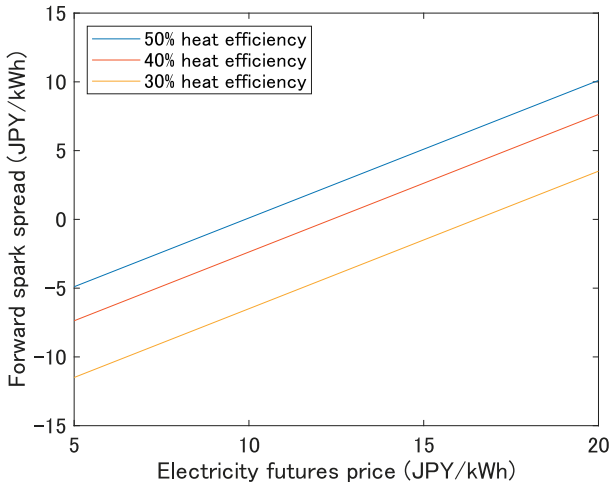


Fig. 2. The correspondence between the electricity futures price and the forward spark spread of each generating unit. The LNG futures price is fixed at 10 (USD/MMBTU).

Since the forward spark spread of each generating unit is different depending on its heat efficiency, in order to display the simulation result of each unit on the same graph, we fix the LNG futures price at 10 (USD/MMBTU) and vary the electricity futures price from 5 to 20 (JPY/kWh) at 0.1 intervals, instead of varying the forward spark spread. The correspondence between the electricity futures price and the forward spark spread of each generating unit is as shown in Fig. 2.

Based on the optimization model described in the previous section, we conduct a simulation of forward spark spread trading. The simulation is performed as a sensitivity analysis with the electricity futures price as the input variable and the optimal futures position, total profit, and number of LNG cargoes to buy/sell as the output variables. For the optimization calculation, we use *intilinprog*, a mixed-integer linear programming solver of Matlab.

In practice, power producers need to perform this optimization calculation for all delivery months in response to the daily changing forward spark spread.

### 5.2 Simulation results

The optimal electricity futures position based on the simulation gives the following results as shown in Fig. 3:

(1) Each unit's position decreases in a step-wise manner against the electricity futures price.
(2) Each unit takes a long (short) position when the forward spark spread is negative (positive), except when the possible futures position $D_i$ or $U_i$ is zero,

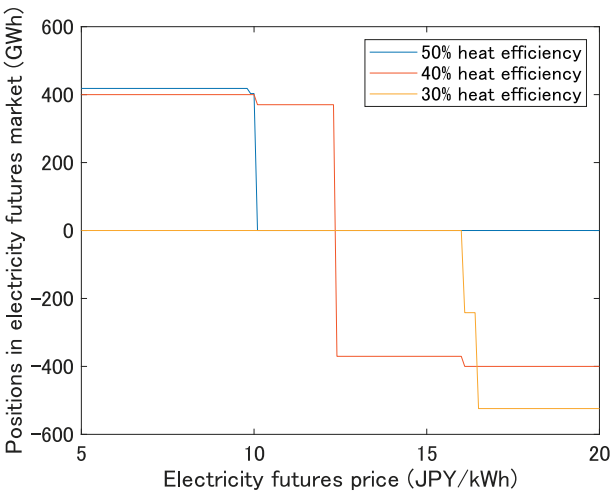


Fig. 3. Optimal electricity futures position of each generating unit.

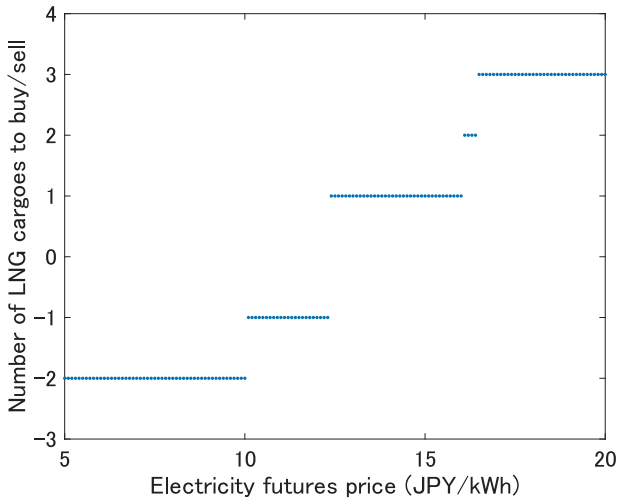


Fig. 4. Number of LNG cargoes to buy/sell in the spot LNG market.

i.e. when there is no additional/reducible electricity in the predetermined power generation plan.

(3) The total futures position of 3 generating units is limited by the constraint on the number of tradable LNG cargoes. Therefore, each unit may not necessarily take the maximum/minimum position. For example, Unit 1 only takes about 420 futures positions, although it can take a maximum of 800. Unit 3 only takes about −520 futures positions, although it can take a minimum of −800.

The number of LNG cargoes to buy/sell in the optimal solution is as shown in Fig. 4. As the electricity futures price increases, the number of LNG cargoes increases in a step-wise manner from $j = -2$ to $j = 3$.

The total profit from forward spark spread trading gives the following results as shown in Fig. 5:

(1) The total profit shows a U-shaped curve against the electricity futures price.

(2) For example, the actual market data in Fig. 1 shows that the 50% heat efficiency unit has a forward spark spread of −5 (JPY/kWh) for the delivery month of May 2024. This situation is similar to the simulation result when the electricity futures price is set to 5 (JPY/kWh) (see Fig. 2). Fig. 5 shows that power producers have an opportunity to make a profit of 5,000 million JPY by trading the forward spark spread.

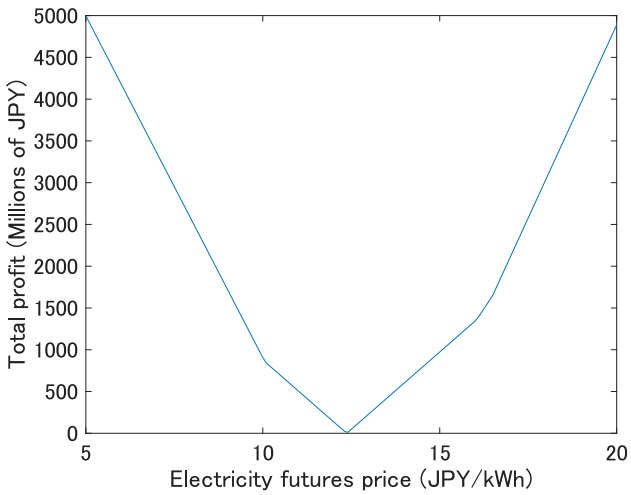


Fig. 5. Total profit from forward spark spread trading.

## 6. Conclusion

In this study, we developed a mixed-integer programming model that maximizes the profit of the gas-fired power generation business in Japan under the practical constraints of LNG cargo size, LNG tank capacity, and multiple generating units with different heat efficiencies.

Using the model, Japanese power producers can calculate the optimal position in the electricity futures market and the number of spot LNG cargoes to buy/sell in response to the daily changing forward spark spread. The results of the numerical simulation pave the way for new opportunities to generate additional profit.

## References

[1] A. Eydeland and K. Wolyniec, Energy and Power Risk Management: New Developments in Modeling, Pricing, and Hedging, John Wiley & Sons, Hoboken, 2003.

[2] M. Endo and T. Hattori, Practices of power and fuel trading and market risk management in Europe: Challenges to strengthen the trading function (in Japanese), Central Research Institute of Electric Power Industry Report Y15003, Central Research Institute of Electric Power Industry, 2015.

[3] M. Endo and T. Hattori, Revenue management of power generation by the power and fuel trading and optimization of asset operation: A case study of German utilities (in Japanese), Central Research Institute of Electric Power Industry Report Y14012, Central Research Institute of Electric Power Industry, 2015.

[4] M. Tsutsui and M. Endo, Factors to reinforce bargaining power in LNG trading and formation of LNG futures market (in Japanese), Central Research Institute of Electric Power Industry Report Y12006, Central Research Institute of Electric Power Industry, 2013.

[5] M. Endo and T. Matsumoto, Consideration for liquidation of the electricity futures market: Realization of strategic risk hedge trading (in Japanese), Rev. Electr. Econ., **67** (2020), 51–63.

[6] T. Shiina, Stochastic programming model for unit commitment problem (in Japanese), Trans. JSIAM, **13** (2003), 181–190.